

Summary Sheet

Pruned evidence-gated pharmacogenomic LLM-alert capstone

Author: Ravi Bajaj | Course: AI in Healthcare

Conference / Symposium

Field	Value
Name	Machine Learning for Health Symposium (ML4H 2026)
URL	https://ml4h.ahli.cc/
Author instructions / CFP	https://ml4h.ahli.cc/submit/call-for-papers/
Submission deadline	September 10, 2026, 11:59 PM AoE
Author notification	October 22, 2026
Camera-ready deadline	November 7, 2026 (tentative)
Event dates	December 6-7, 2026
Location	Sydney, Australia
Suggested track	Findings Track for preliminary specification-conformance work; Proceedings Track only after independent review.

Title and Abstract

Title: Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims: Detecting Overclaiming Relative to Guideline-Supported Evidence

Abstract: This capstone studies a narrowed clinical AI safety problem: LLM-drafted pharmacogenomic and genomic medication alerts can overstate what guideline-supported evidence actually permits. I implement a synthetic pre-display evidence gate with three checks: citation/guideline support, population fit, and endpoint/actionability. Stage 1 does not call an LLM or extract evidence features from free text; it tests a deterministic controller over structured evidence annotations. In 33 author-designed synthetic cases, the controller blocked or narrowed all 23 designed overclaim archetypes, allowed all 10 bounded alerts, and produced no inappropriate denials. Action conformance was 33/33 and primary-gate conformance was 31/33, with PGX19 and PGX24 revealing citation-first precedence effects. One-gate ablations show that each gate uniquely catches at least one failure archetype. These counts show specification conformance, not clinical utility, independent safety, or generalization.

Generative AI Use

Generative AI tools were used for brainstorming, critique, code scaffolding, drafting, packaging, and formatting. The author selected the final scope, directed the pruning, verified sources, reviewed generated text, ran local tests, and remains responsible for all claims, limitations, and submission materials.

Optional Data, Code, and Tools

- Synthetic 33-case pharmacogenomic/genomic medication-alert case bank.
- Executable three-gate controller and tests.
- One-gate ablation CSV showing isolated necessity of citation/guideline support, population fit, and endpoint/actionability gates.
- CSV/JSON results for overclaim detection, inappropriate denial, calibration, and gate/error categories.

- No real patient data, no protected health information, no diagnosis, and no treatment recommendation.